**3.** A biased spinner can land on the numbers 1, 2, 3, 4 or 5 with the following probabilities.

| **Number on spinner** | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| **Probability** | 0.3 | 0.1 | 0.2 | 0.1 | 0.3 |

The spinner will be spun 80 times and the mean of the numbers it lands on will be calculated.
Find an estimate of the probability that this mean will be greater than 3.25

**(6)**

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7. A spinner can land on red or blue. When the spinner is spun, there is a probability of $\frac{1}{3}$ that it lands on blue. The spinner is spun repeatedly.

The random variable $B$ represents the number of the spin when the spinner first lands on blue.

(a) Find (i) $P(B = 4)$

(ii) $P(B \leqslant 5)$

**(4)**

(b) Find $E(B^2)$

**(3)**

Steve invites Tamara to play a game with this spinner.

Tamara must choose a colour, either red or blue.

Steve will spin the spinner repeatedly until the spinner first lands on the colour Tamara has chosen. The random variable $X$ represents the number of the spin when this occurs.

If Tamara chooses red, her score is $e^X$

If Tamara chooses blue, her score is $X^2$

(c) State, giving your reasons and showing any calculations you have made, which colour you would recommend that Tamara chooses.

**(5)**

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**4.** The discrete random variable $X$ has the following probability distribution.

| $x$ | $-5$ | $-2$ | $3$ | $4$ |
|---|---|---|---|---|
| $P(X = x)$ | $\frac{1}{12}$ | $\frac{1}{6}$ | $\frac{1}{4}$ | $\frac{1}{2}$ |

(a) Find $\text{Var}(X)$
**(3)**

The discrete random variable $Y$ is defined in terms of the discrete random variable $X$

When $X$ is negative, $Y = X^2$

When $X$ is positive, $Y = 3X - 2$

(b) Find $P(Y < 9)$
**(3)**

(c) Find $E(XY)$
**(2)**

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**4.** Members of a photographic group may enter a maximum of 5 photographs into a members only competition.
Past experience has shown that the number of photographs, $N$, entered by a member follows the probability distribution shown below.

| $n$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $P(N = n)$ | $a$ | 0.2 | 0.05 | 0.25 | $b$ | $c$ |

Given that $E(4N + 2) = 14.8$ and $P(N = 5 \mid N > 2) = \frac{1}{2}$

(a) show that $\text{Var}(N) = 2.76$

**(6)**

The group decided to charge a 50p entry fee for the first photograph entered and then 20p for each extra photograph entered into the competition up to a maximum of £1 per person. Thus a member who enters 3 photographs pays 90p and a member who enters 4 or 5 photographs just pays £1

Assuming that the probability distribution for the number of photographs entered by a member is unchanged,

(b) calculate the expected entry fee per member.

**(3)**

Bai suggests that, as the mean and variance are close, a Poisson distribution could be used to model the number of photographs entered by a member next year.

(c) State a limitation of the Poisson distribution in this case.

**(1)**

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**2.** The discrete random variable $X$ has probability distribution

| $x$ | $-5$ | $-1$ | $0$ | $5$ | $b$ |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.3 | 0.25 | 0.1 | 0.15 | 0.2 |

where $b$ is a constant and $b > 5$

(a) Find $E(X)$ in terms of $b$

**(1)**

Given that $\text{Var}(X) = 34.26$

(b) find the value of $b$

**(4)**

(c) Find $P(X^2 < 2 - 3X)$

**(4)**

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**1.** The discrete random variable $X$ has probability distribution

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $3$ |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.25 | $a$ | $b$ | $a$ | 0.30 |

where $a$ and $b$ are probabilities.

(a) Find $E(X)$

**(2)**

Given that $\text{Var}(X) = 3.9$

(b) find the value of $a$ and the value of $b$

**(4)**

The independent random variables $X_1$ and $X_2$ each have the same distribution as $X$

(c) Find $P(X_1 + X_2 > 3)$

**(3)**

**1.** The discrete random variable $X$ has the following probability distribution

| $x$ | $-1$ | 0 | 1 | 3 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.2 | 0.1 | 0.2 | 0.25 | 0.25 |

(a) Find $\mathrm{Var}(X)$

**(3)**

(b) Find $\mathrm{Var}(X^2)$

**(3)**

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